

Mishika Lunia

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Education

Indian Institute of Science Education and Research, Bhopal, BS in Electrical Engineering and Computer Science (GPA: 7.88/10.0)

May 2023

Experience

Software Engineer, Orange Data Tech

Aug 2025 – Present

- Architected and deployed a production-grade **Adaptive Iterative RAG** for legal compliance platform, integrating multimodal document understanding.
- Built an automated remittance reconciliation platform using **Azure Service Bus**, Azure Blob Storage, Power Automate, and LLM pipelines, achieving **98% document extraction accuracy** and removing manual invoice-payment matching effort.
- Developed full-stack features for a B2B fintech SaaS using **React, Django, REST APIs, and PostgreSQL** and a client-facing product demonstration portal which led to onboarding of new clients.
- Enhanced and integrated the **Papermerge** document management system into an enterprise payroll platform by implementing **AES-256 encryption**, Redis-based rate limiting, and role-based access controls.

Research Assistant, DoMS, IIT Roorkee (Guided by: Dr. Gaurav Dixit)

May 2024 – Apr 2025

Research Focus: Impact of Government IT/ITeS Policies on Corporate Profitability in Indian Markets

- Conducted panel data analysis across 105,000+ firm-year observations (2010–2023) to evaluate the causal impact of Indian IT/ITeS policies (**Digital India, IndiaStack/UPI, Make in India**, etc.) on firm-level profitability across Fintech, IT, Manufacturing, and Transport sectors.
- Designed and implemented **Two-Way Fixed Effects Difference-in-Differences (TWFE-DiD)** models with event study validation (± 4 -year windows), **Arellano–Bond Difference GMM**, and dynamic panel robustness checks across four sector-specific samples; validated all specifications using Hausman, Breusch–Pagan, Sargan, and AR(1)/AR(2) diagnostic tests.
- Applied **XGBoost**, Random Forest, stacked ensemble, LSTM, and GRU models for firm profitability prediction test ($R^2 = 0.83$, **MDA = 80%**); found income/expense ratio and firm size to be the strongest predictors of profitability, with IT investment variables consistently exhibiting mid-to-low feature importance.
- Found no statistically significant causal effect of the policies studied on firm profitability across all econometric estimators once firm-level heterogeneity was controlled.

Certifications

Executive Postgraduate Program (Online), IIT Roorkee

Jun 2023 – May 2024

Data Science, AI/ML & Data Engineering,

Technologies

Languages: Python, R, C++, Javascript, MySQL

Technologies: PyTorch, HuggingFace, Docker, Rest-API, FastAPI, Langchain, Django, Azure, Azure AI Foundry, Azure Data Studio, Linux